Serverless Architectures

OK, NO MORE SERVERS – NOW WHAT?

Gregor Hohpe

Sr. Principal Evangelist
AWS

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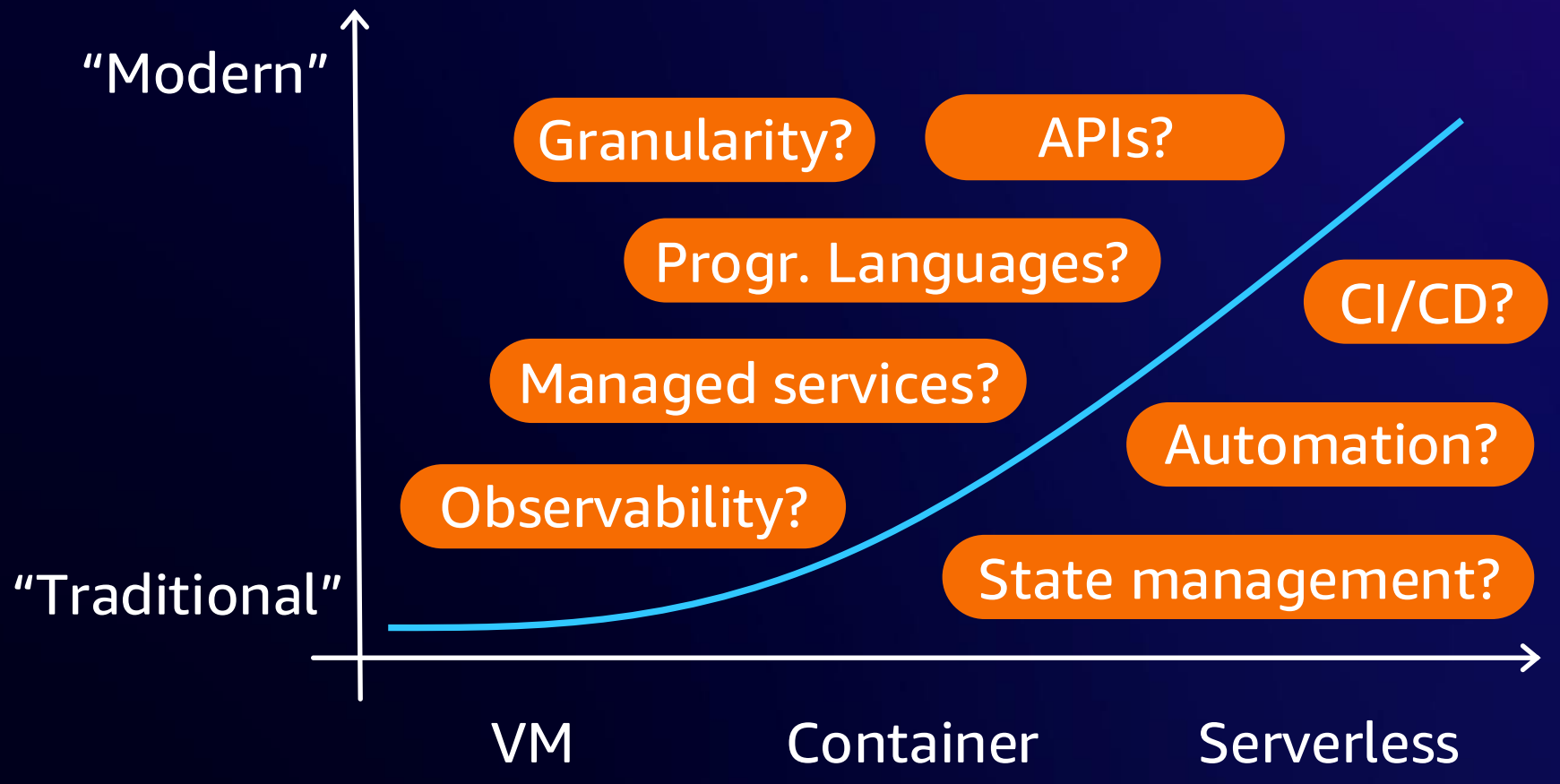
How modern is your cloud application?



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Serverless – Not just a run-time

COMPUTE

AWS Lambda

AWS Fargate

DATA STORES

Amazon S3

Amazon Aurora Serverless

Amazon DynamoDB

INTEGRATION

Amazon EventBridge

Amazon API Gateway

Amazon SQS

Amazon SNS

AWS Step Functions

AWS AppSync

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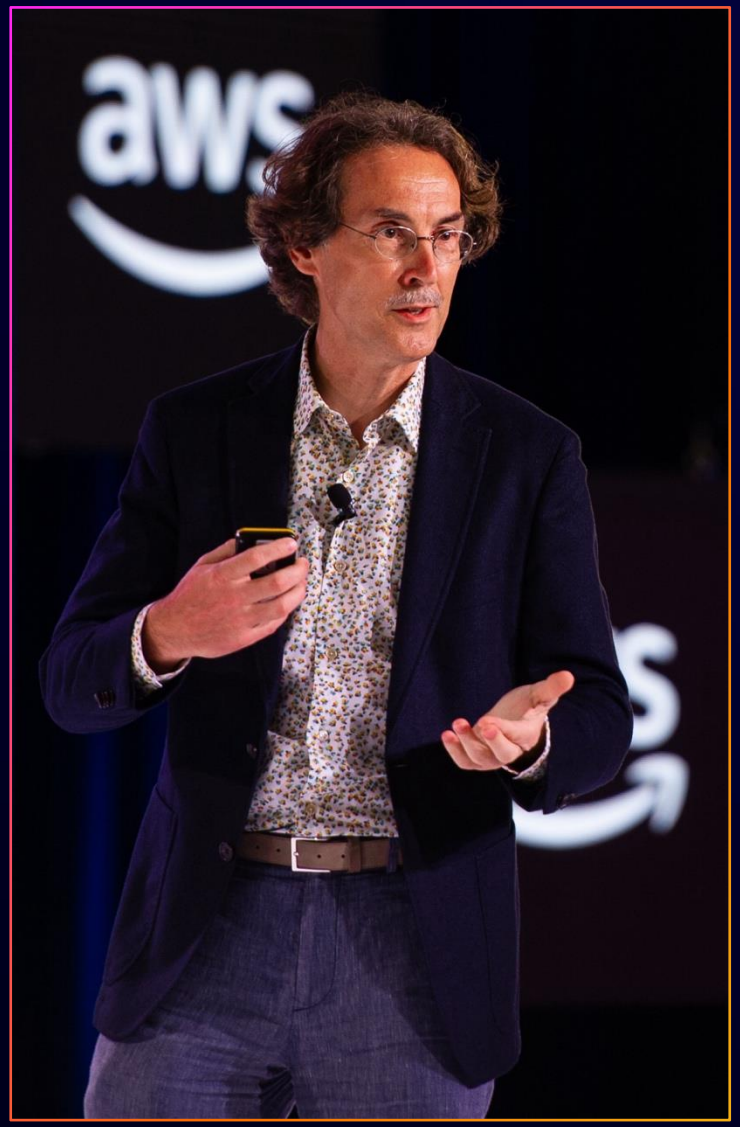

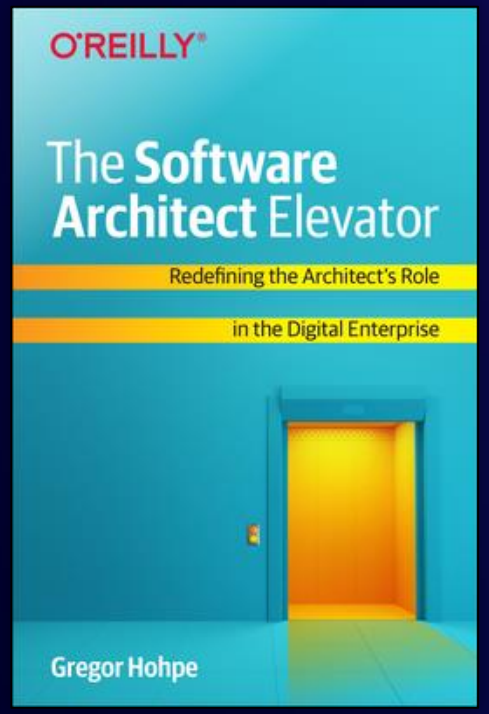

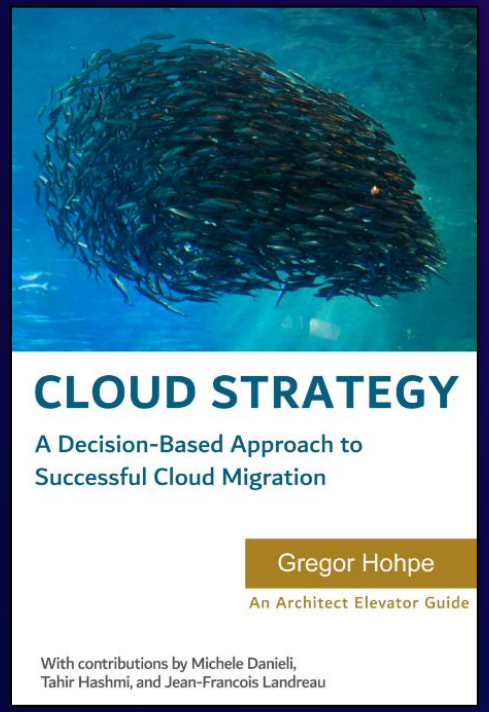

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Serverless Architectures: Under the Lambda covers

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AWS Lambda – Launched Nov 2014

- Provide your code or image; we run it as an event when things happen
- Pay by the millisecond, up to 15mins
- Package as .zip file (250 MB) or container image (10 GB)
- Java, Go, Node.js, .NET, Python, and Ruby, or bring-your own
- No provisioning or managing servers
- Scale up in milliseconds in response to traffic

Lambda is the compression algorithm for experience

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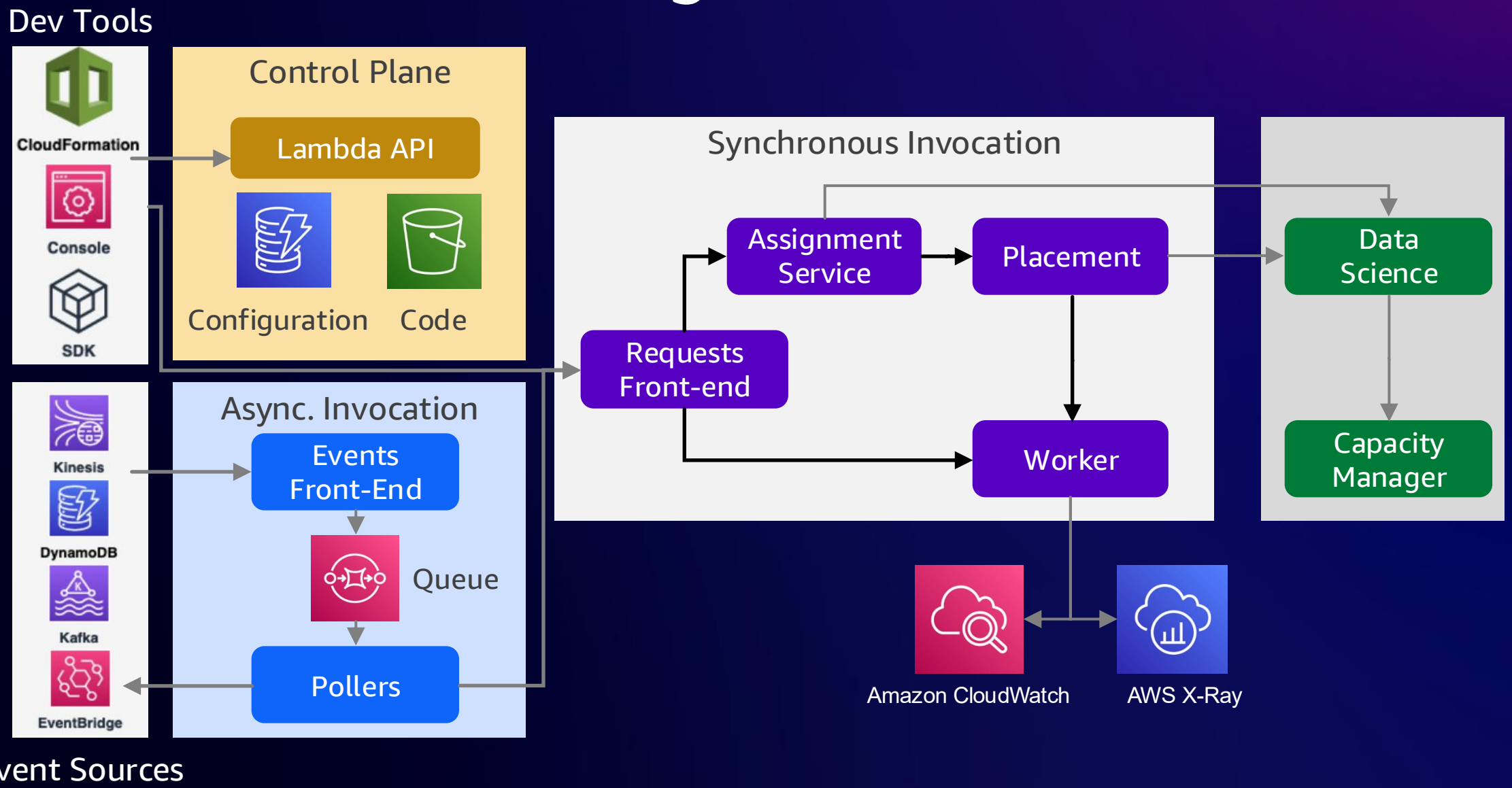
Lambda Architecture – High-level View



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Our experience: Implementation considerations

- Placement (Assignment Service)
  - Fault tolerant, leader-follower model
- Worker manager
- Isolation levels
  - Firecracker micro-VMs
  - Tenant de-correlation
- Initialization / cold starts
  - SnapStart
  - Chunks, read-ahead

https://www.infoq.com/news/2023/10/aws-lambda-under-the-hood/

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Serverless Architectures: Of boxes and lines

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Serverless applications are distributed and fine-grained.

Distribute responsibly!

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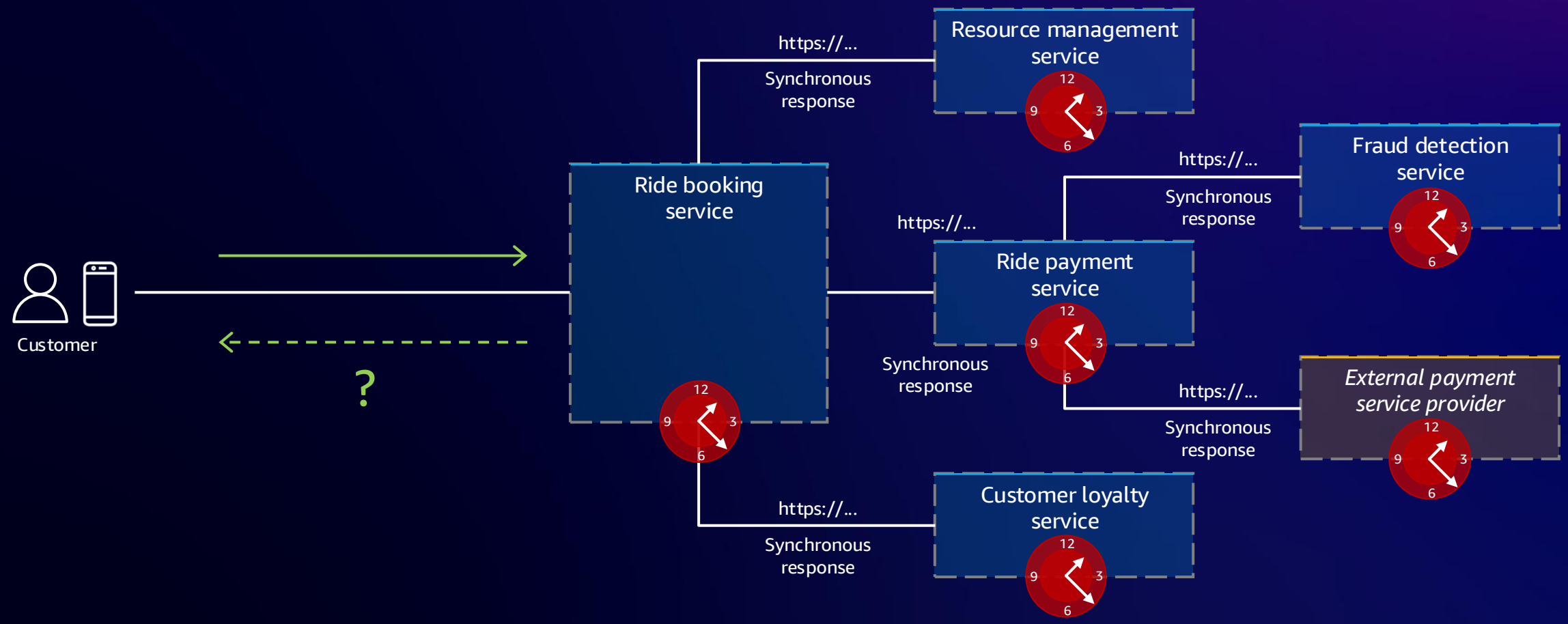
Friends don’t let friends use synchronous integration



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• Event-driven architectures

- Lambda has synchronous and asynchronous invocation models
- Push models, pull models, streaming
- Serverless integration provides orchestration, event routing, and message queues

Distributed, event-driven architecture

Reduced Dependencies	Independent Scalability	Failure Isolation	Evolvable Architecture

aws

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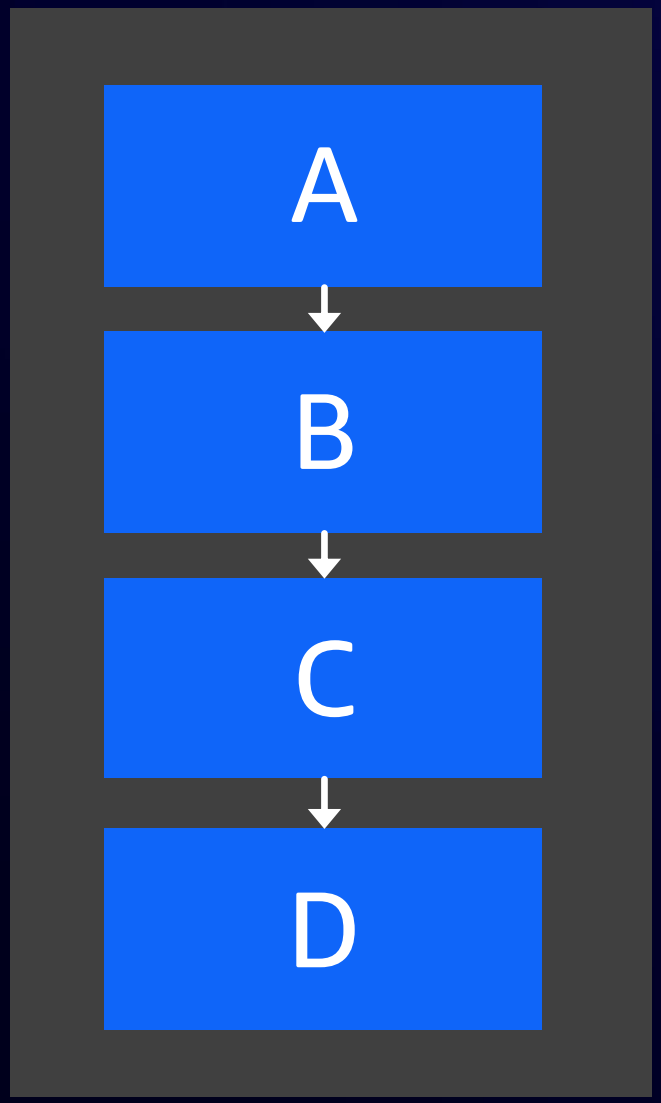
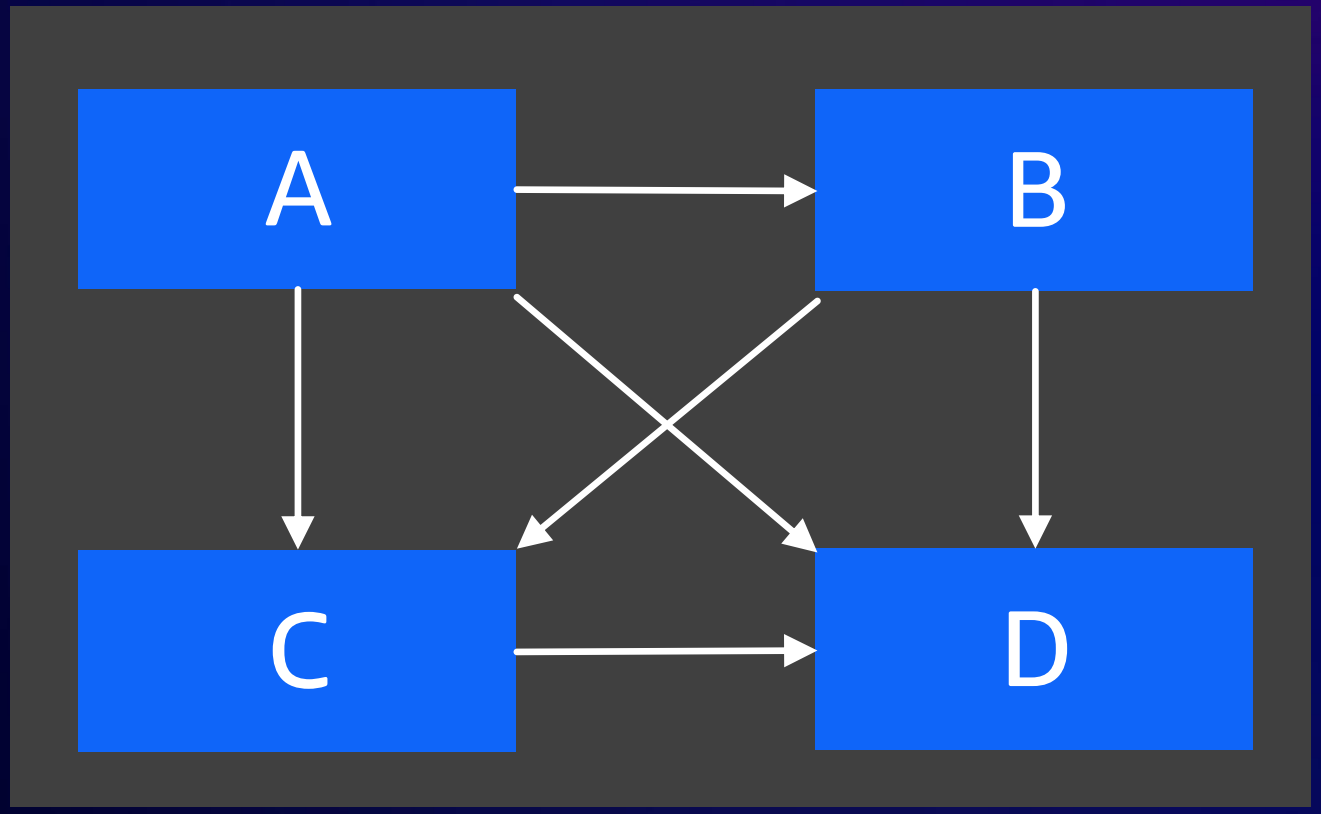
Behold the lines!





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Connecting two systems. How hard can it be?

Data or control flow?

Pub-sub or point-to-point?

Interaction model?

Sync/async?

Polling?

Data format?

Distributed?

Conversation State?

Systems or instances?

Error handling?

Idempotency?

Partial failures?

Retries?

Back-off?

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Control flow vs. data flow

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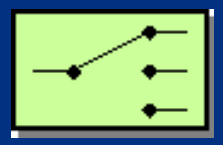
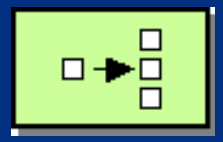
Enterprise Integration Patterns

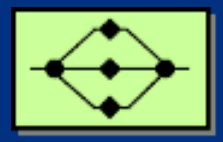
Data (Message) Flow

Message Routing

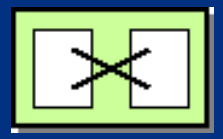

Content-based Router

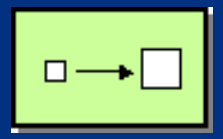
Message Filter

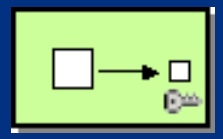

Splitter


Scatter Gather

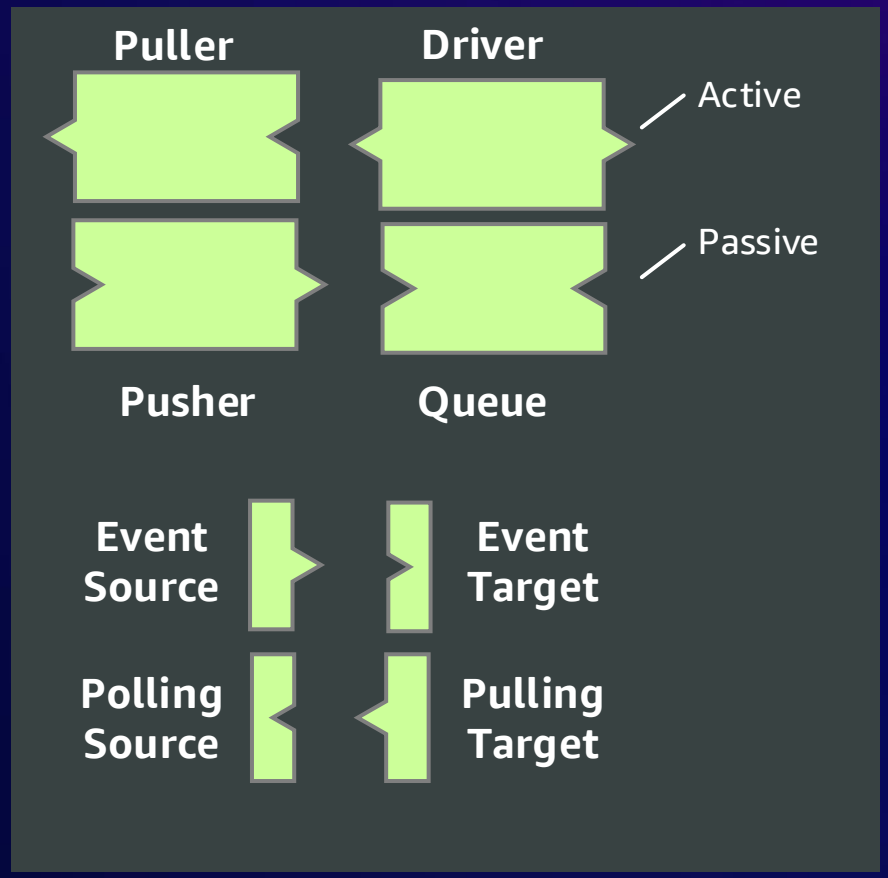
Message Transformation


Message Translator


Content Enricher


Claim Check

Control Flow



Source: EnterpriseIntegrationPatterns.com

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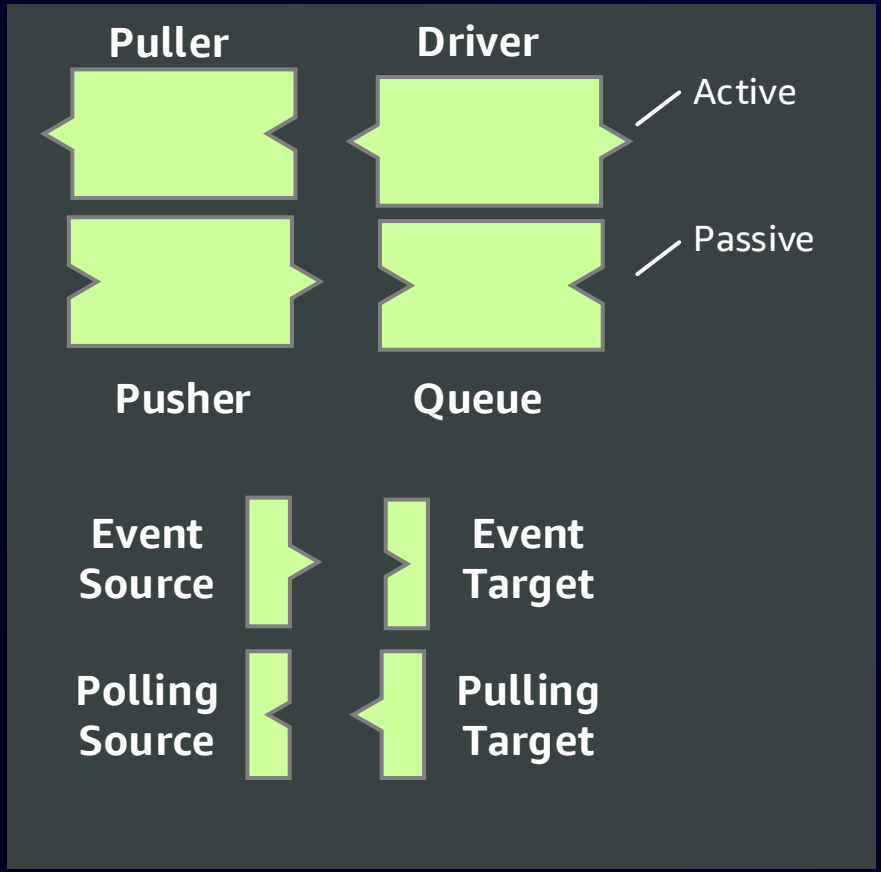
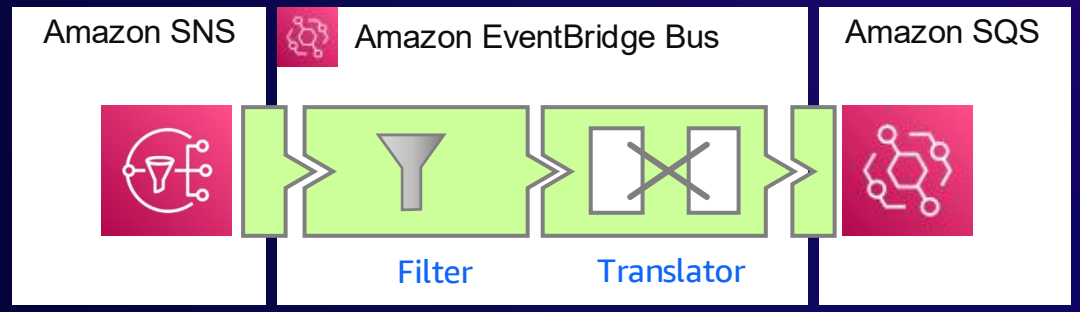
Visualizing control flow: EventBridge Bus

Control Flow

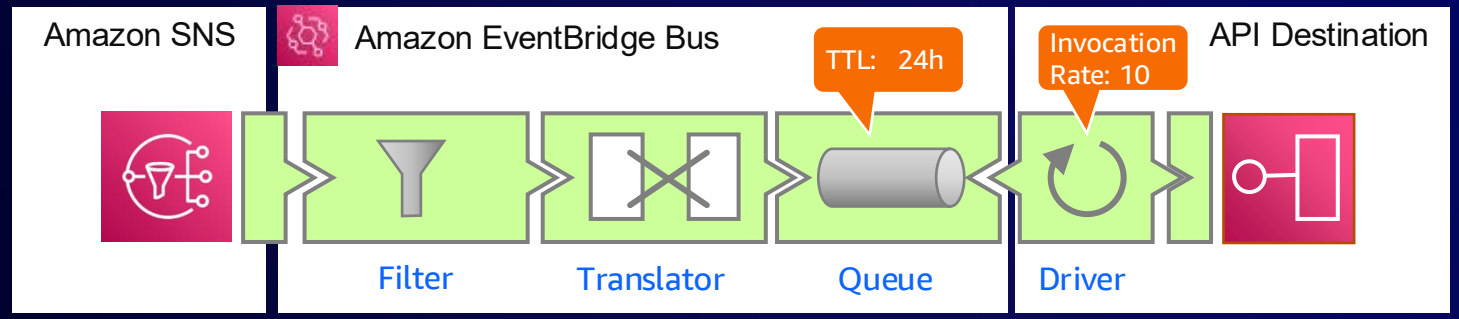


Source: EnterpriseIntegrationPatterns.com

EventBridge with Event Target



EventBridge with Rate Limited Target



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Control flow is essential to understanding dynamic system behavior such as latency, scaling, batching, time-outs

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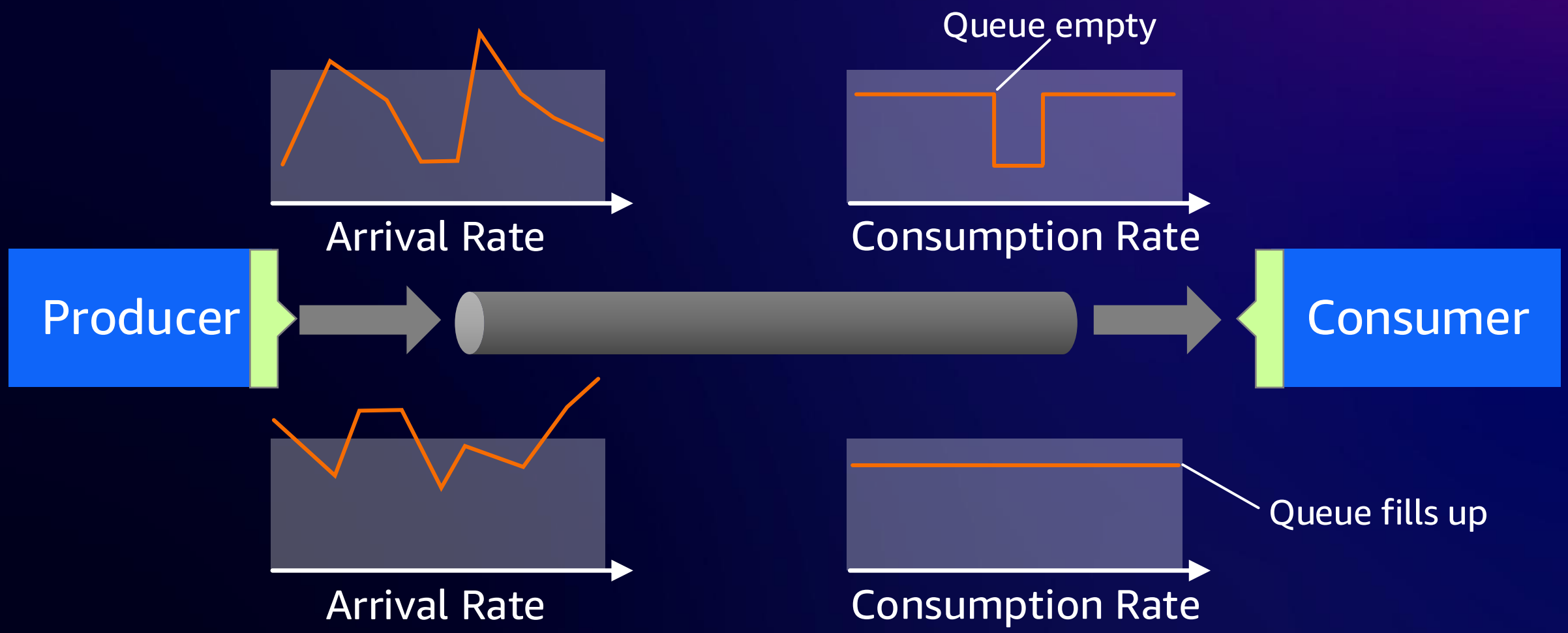
Queues invert flow control, shape traffic



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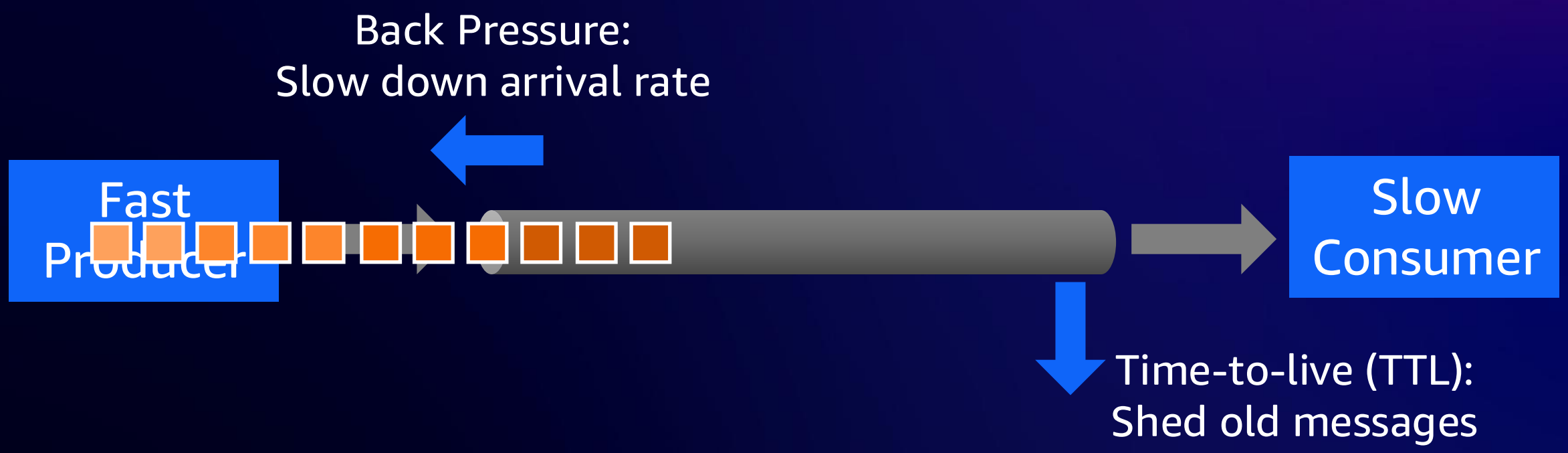
Queues require flow control



TTL is an SQS setting, backpressure has to be built.

Even valuable messages have a meaningful time to live.

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Queues decouple control flow but require flow control.

Hard-earned experience from an integration architect

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Serverless Architectures: Coordination

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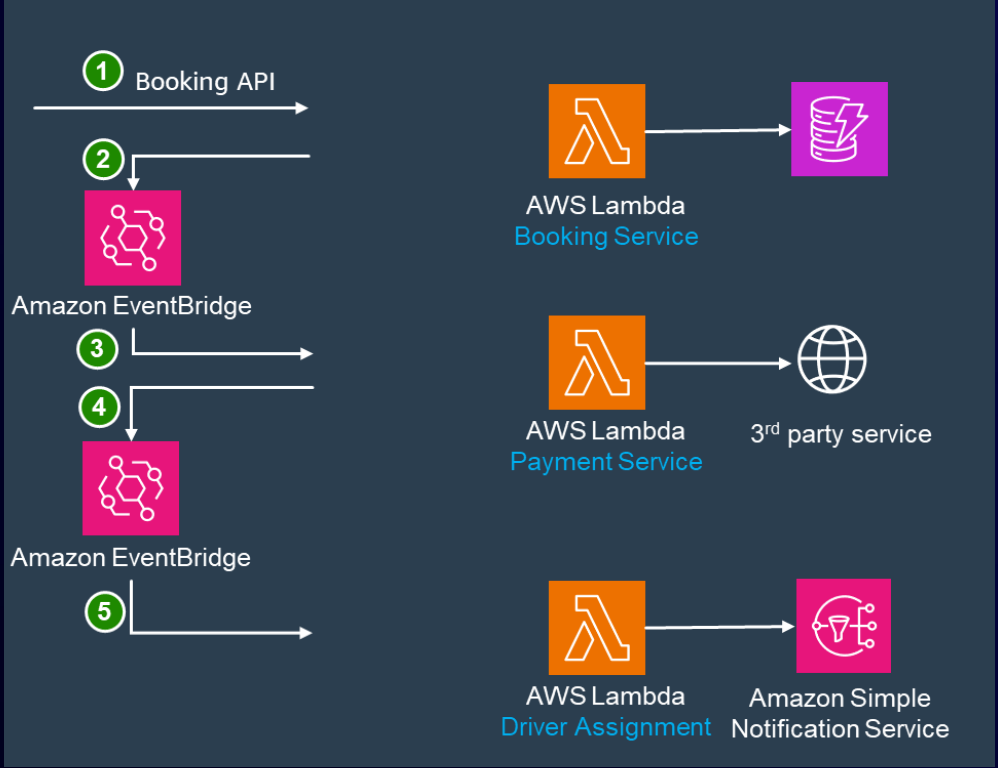
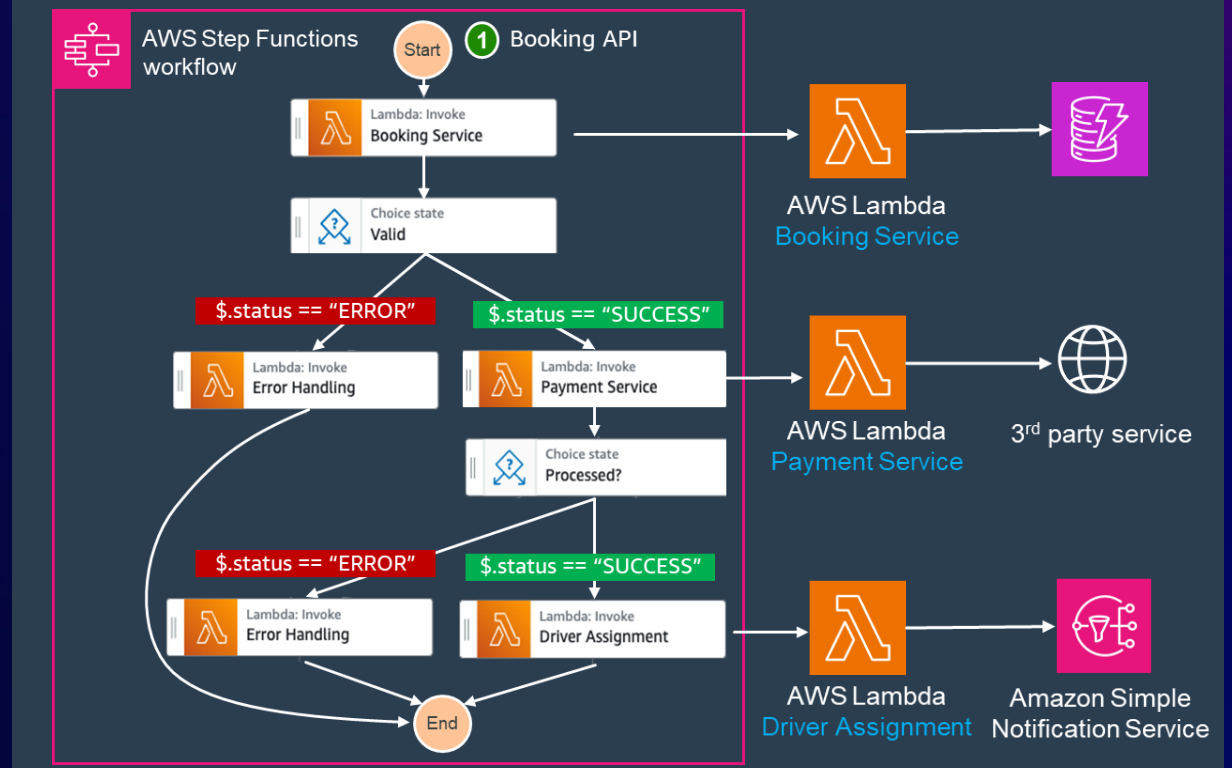
Choreography vs. Orchestration

Original Design



- ✓ Business logic can evolve easily (add, update services)
- ✓ Services can scale independently
- ➢ Lambda must publish events
- ✗ Transactional consistency across services is difficult
- ✗ End-to-end monitoring ,handling errors, and timeout is difficult

Refactored Design



- ✓ Centralized business logic in Orchestrator
- ✓ Transaction management across services
- ✓ End-to-end monitoring and error handling easier
- ✗ Tightly coupled
- ✗ Orchestrator can become single point of failure / limitation
- ✗ Single point of maintenance (can be good or bad)

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Serverless Architectures: Automation

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Cloud
− Automation
= Just another data center

An opinionated architect

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Automation focused on infrastructure ("IaC")

- Object-oriented or functional
- Powerful languages
- Type systems
- Automated testing, TDD, IDE

"Build"

Development teams

- Functionality, UX
- Application architecture
- Write code

Application

"Run"

Infra / Ops teams

- Non-functional "-ilities"
- Sizing, procurement
- Click-Ops -> Write scripts

Automation

- Resource-oriented
- Prefer simple syntax (YAML)
- Strings and integers
- Text editor

https://architectelevator.com/cloud/iac-ifc-trends/

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As application architecture changes, so does the role of cloud automation.

Limiting cloud automation to infrastructure is no longer adequate.

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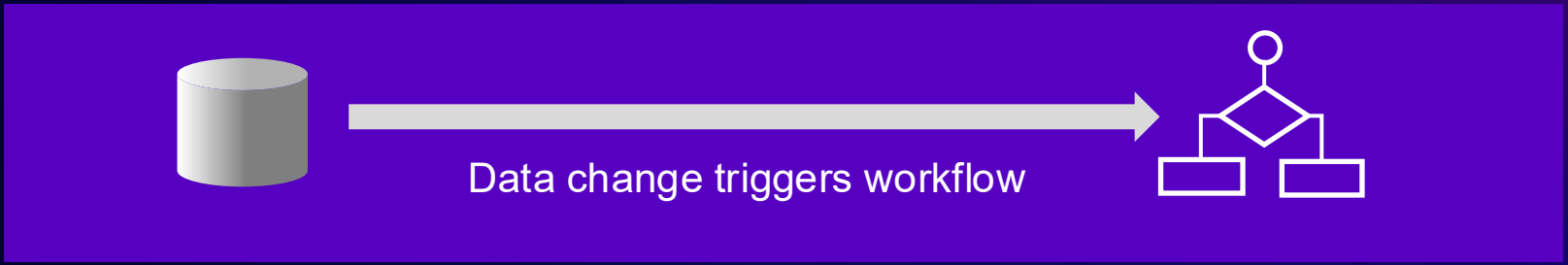
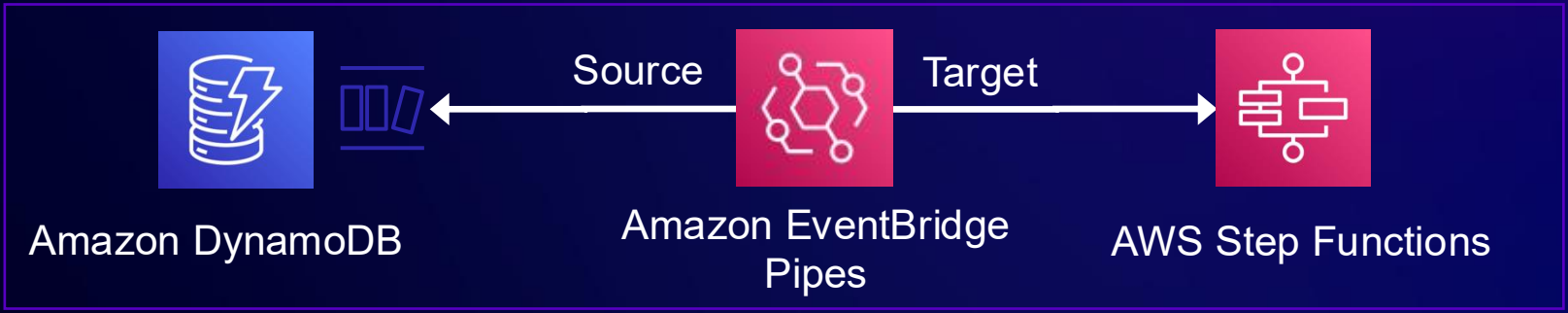
Composition with cloud automation

Architecture

Implementation (Services)

Automation Code






```
this.sourceArn = table.tableStreamArn;
this.sourceParameters = {
  dynamoDbStreamParameters: {
    startingPosition: LATEST,
    maximumBatchingWindowInSeconds: 5,
    maximumRetryAttempts: 10
  },
};
```

Object-Oriented (CDK)

```
Type: AWS::Pipes::Pipe
Properties:
  Description: String
  Name: String
  RoleArn: String
  Source: String
  SourceParameters:
    PipeSourceParameters
```

YAML (CloudFormation)

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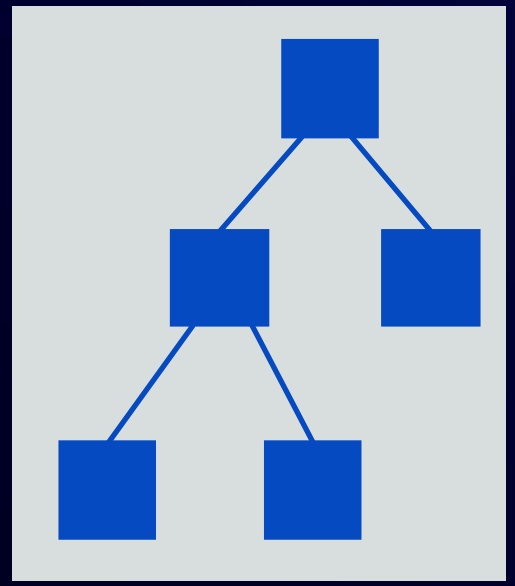
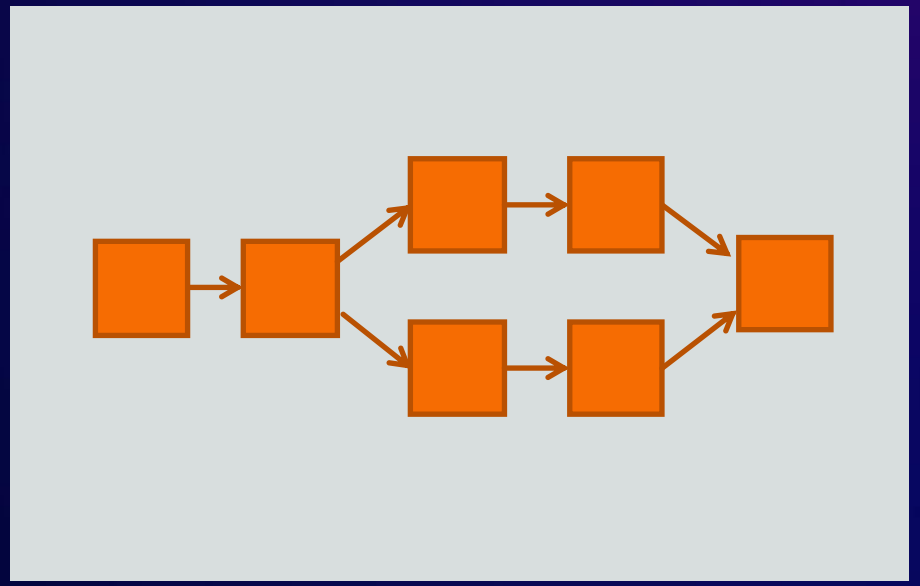
From IaC to AaC: Architecture as Code



Classic IaC (incl. CDK):
Resource hierarchy

Well suited for
infrastructure resources



Serverless Application Architecture:
Data and control flow

Well suited for fine-grained and
loosely coupled applications

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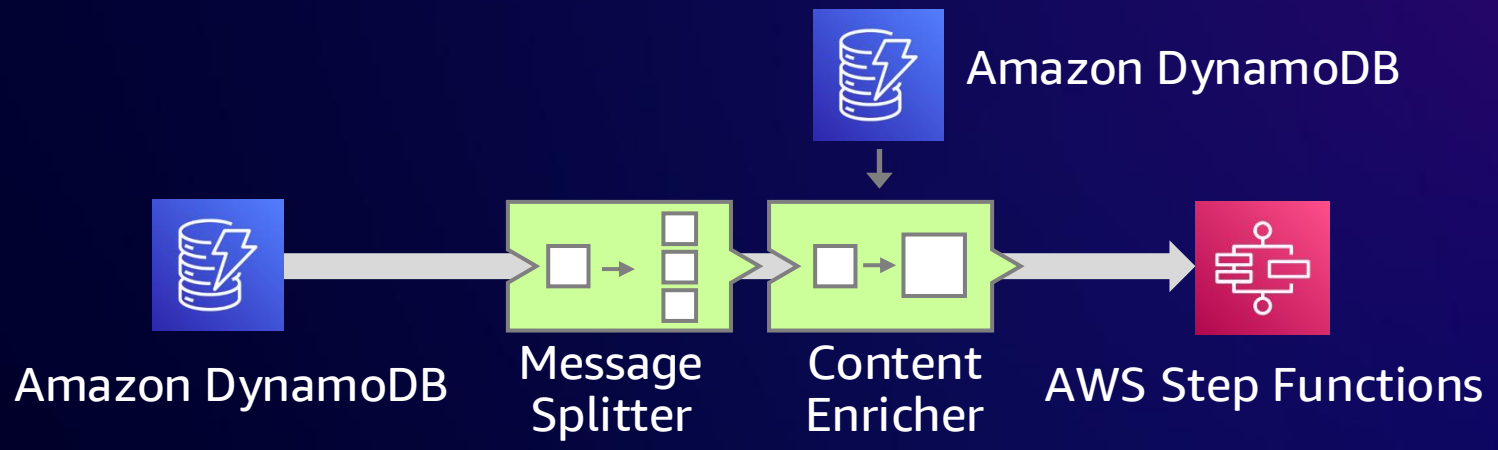
Source: architectelevator.com/cloud/iac-ifc-trends/

Coding in 202x: Fluent APIs, Pipes-and-Filters

Architecture



Automation Code

Implementation with CDK (Experimental)

```
const pipeline = new PipelineBuilder<IPipelineSource, Filter, Channel, IPipelineTarget>()
  .fromSource(new DynamoDBStreamSource(ordersTable))
  .with(new MessageSplitter('$.tickets', ['$.id', '$.userId'], 'common_'))
  .with(new ContentEnricher(lookupTable, { "id": "id" }, "detail"))
  .toTarget(new StepFunctionsTarget(…);

deployPipe(this, "myPipeline", pipeline, LogLevel.ALL);
```

Implementation

Icons: EnterpriseIntegrationPatterns.com

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Program to your architecture, not cloud services

Traditional Model

Architecture

Manual Translation

Implementation

Manual Translation

Automation Code

Cloud Compiler Approach

Architecture

Direct Mapping

Automation Code

Automatic Translation

Implementation

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Next-gen Serverless Automation:

Actual

~~Infrastructure~~ **as Code**

Application Architecture

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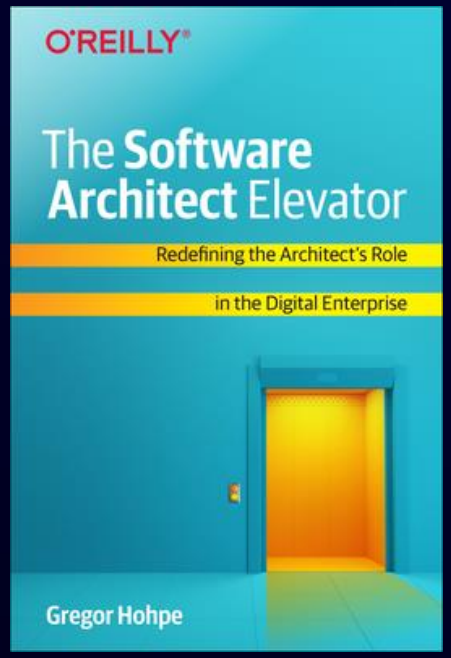
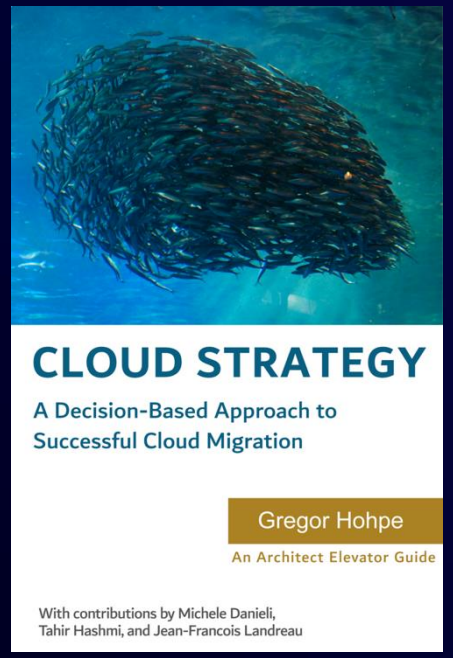
Want more?





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